

1 Introduction

Difference-in-Differences (DiD) has become one of the most popular research designs used to evaluate causal effects of policy interventions. In its canonical format, there are two time periods and two groups: in the first period no one is treated, and in the second period some units are treated (the treated group), and some units are not (the comparison group). If, in the absence of treatment, the average outcomes for treated and comparison groups would have followed parallel paths over time (which is the so-called parallel trends assumption), one can estimate the average treatment effect for the treated subpopulation (ATT) by comparing the average change in outcomes experienced by the treated group to the average change in outcomes experienced by the comparison group. Methodological extensions of DiD methods often focus on this standard two periods, two groups setup; see, e.g., [Heckman et al. \(1997, 1998\)](#), [Abadie \(2005\)](#), [Athey and Imbens \(2006\)](#), [Qin and Zhang \(2008\)](#), [Bonhomme and Sauder \(2011\)](#), [de Chaisemartin and D’Haultfœuille \(2017\)](#), [Botosaru and Gutierrez \(2018\)](#), [Callaway et al. \(2018\)](#), and [Sant’Anna and Zhao \(2020\)](#).¹

Many DiD empirical applications, however, deviate from the canonical DiD setup and have more than two time periods and variation in treatment timing. In this article, we provide a unified framework for average treatment effects in DiD setups with multiple time periods, variation in treatment timing, and when the parallel trends assumption holds potentially only after conditioning on observed covariates. We concentrate our attention on DiD with staggered adoption, i.e., to DiD setups such that once units are treated, they remain treated in the following periods.

The core of our proposal relies on separating the DiD analysis into three separate steps: (i) identification of policy-relevant disaggregated causal parameters; (ii) aggregation of these parameters to form summary measures of the causal effects; and (iii) estimation and inference about these different target parameters. Our approach allows for estimation and inference on interpretable causal parameters allowing for arbitrary treatment effect heterogeneity and dynamic effects, thereby completely avoiding the issues of interpreting results of standard two-way fixed effects (TWFE) regressions as causal effects in DiD setups as pointed out by [Borusyak and Jaravel \(2017\)](#), [de Chaisemartin and D’Haultfœuille \(2020\)](#), [Goodman-Bacon \(2019\)](#), [Sun and Abraham \(2020\)](#), and [Athey and Imbens \(2018\)](#). In addition, it adds transparency and objectivity to the analysis ([Rubin \(2007, 2008\)](#)), and allows researchers to exploit a variety of estimation methods to answer different questions of interest.

The identification step of the analysis provides a blueprint for the other steps. In this paper, we pay particular attention to the disaggregated causal parameter that we call the *group-time average treatment effect*, i.e., the average treatment effect for group g at time t , where a “group” is defined by the time period when units are first treated. In the canonical DiD setup with two periods and two groups, these parameters reduce to the ATT which is typically the parameter of interest in that setup. An attractive feature of the group-time average treatment effect parameters is that they do not directly restrict heterogeneity with respect to observed covariates, the period in which units are first treated, or the evolution of treatment effects over time. As a consequence, these easy-to-interpret causal parameters can be directly used for learning about treatment effect heterogeneity, and/or to construct many other more aggregated causal

¹See Section 6 of [Athey and Imbens \(2006\)](#) and Theorem S1 in [de Chaisemartin and D’Haultfœuille \(2017\)](#) for notable exceptions that cover multiple periods and multiple groups.

adoption in the literature. We interpret this assumption as if units do not “forget” about the treatment experience.⁴

Define G as the time period when a unit first becomes treated. Under Assumption 1, for all units that eventually participate in the treatment, G defines which “group” they belong to. If a unit does not participate in any time period, we arbitrarily set $G = \infty$. We define G_g to be a binary variable that is equal to one if a unit is first treated in period g (i.e., $G_{i,g} = \mathbf{1}\{G_i = g\}$) and define C to be a binary variable that is equal to one for units that do not participate in the treatment in any time period (i.e., $C_i = \mathbf{1}\{G_i = \infty\} = 1 - D_{i,\mathcal{T}}$). Let $\bar{g} = \max_{i=1,\dots,n} G_i$ be the maximum G in the dataset. Next, denote the generalized propensity score as $p_{g,s}(X) = P(G_g = 1 | X, G_g + (1 - D_s)(1 - G_g) = 1)$. Note that $p_{g,s}(X)$ indicates the probability of being first treated at time g , conditional on pre-treatment covariates X and on either being a member of group g (in this case, $G_g = 1$) or a member of the “not-yet-treated” group by time s (in this case, $(1 - D_s)(1 - G_g) = 1$). Many of our results use a specialized version of this generalized propensity score, and, henceforth, we define $p_g(X) = p_{g,\mathcal{T}}(X) = P(G_g = 1 | X, G_g + C = 1)$ which is the probability of being first treated in period g conditional on covariates and either being a member of group g or not participating in the treatment in any time period. Let $\mathcal{G} = \text{supp}(G) \setminus \{\bar{g}\} \subseteq \{2, 3, \dots, \mathcal{T}\}$ denote the support of G excluding $\bar{g}$.⁵ Likewise, let $\mathcal{X} = \text{supp}(X) \subseteq \mathbb{R}^k$ denote the support of the pre-treatment covariates. Finally, for a generic $\delta \geq 0$, let $\mathcal{G}_\delta = \mathcal{G} \cap \{2 + \delta, 3 + \delta, \dots, \mathcal{T}\}$.

Next, we set up the potential outcomes framework. Here, we combine the dynamic potential outcomes framework of Robins (1986, 1987) with the multi-stage treatment adoption setup discussed by Heckman et al. (2016); see also Sianesi (2004). Let $Y_{i,t}(0)$ denote unit i ’s untreated potential outcome at time t if they remain untreated through time period $\mathcal{T}$; i.e., if they were not to participate in the treatment across all available time periods. For $g = 2, \dots, \mathcal{T}$, let $Y_{i,t}(g)$ denote the potential outcome that unit i would experience at time t if they were to first become treated in time period g . Note that our potential outcomes notation accounts for potential dynamic treatment selection, though it also accommodates (pre-specified) treatment regimes (Murphy et al., 2001; Murphy, 2003). The observed and potential outcomes for each unit i are related through

$$Y_{i,t} = Y_{i,t}(0) + \sum_{g=2}^{\mathcal{T}} (Y_{i,t}(g) - Y_{i,t}(0)) \cdot G_{i,g} \quad (2.1)$$

In other words, we only observe one potential outcome path for each unit. For those that do not participate in the treatment in any time period, observed outcomes are untreated potential outcomes in all periods. For units that do participate in the treatment, observed outcomes are the unit-specific potential outcomes corresponding to the particular time period when that unit adopts the treatment.

We also impose the following random sampling assumption.

Assumption 2 (Random Sampling). $\{Y_{i,1}, Y_{i,2}, \dots, Y_{i,\mathcal{T}}, X_i, D_{i,1}, D_{i,2}, \dots, D_{i,\mathcal{T}}\}_{i=1}^n$ is independent and identically distributed (*iid*).

Assumption 2 implies that we have access to panel data; our results extend essentially immediately

⁴See Han (2020), de Chaisemartin and D’Haultfoeulle (2020) and Bojinov et al. (2020) for alternative setups where treatment can “turn off”.

⁵When there is a “never treated” set of units with $G = \infty$, $\mathcal{G}$ only excludes this group. When such “never-treated” group is not available, we exclude the latest-treated group as there will be no valid untreated comparison group for them.

treatment at different lengths of exposure to the treatment.

Despite the popularity of these specifications, recent research has shown that one must be very careful in attaching a causal interpretation to these aggregated parameters. For instance, [Borusyak and Jaravel \(2017\)](#), [Goodman-Bacon \(2019\)](#), [de Chaisemartin and D’Haultfoeuille \(2020\)](#), and [Athey and Imbens \(2018\)](#) have shown that, in general, β recovers a weighted average of some underlying treatment effect parameters but some of the weights on these parameters can be negative. This can potentially lead to particularly problematic cases such as the effect of the treatment being positive for all units, but the TWFE estimation resulting in estimates of β that are negative. Even in cases where the weights are not negative, the weights on underlying treatment effect parameters are entirely driven by the TWFE estimation strategy and are sensitive to the size of each group, the timing of treatment, and the total number of time periods (see Theorem 1 in [Goodman-Bacon \(2019\)](#)). The results in this section can be used in exactly the same setup to identify a single interpretable average treatment effect parameter and, thus, provide a way to circumvent the issues with the more common approach.

As discussed by [Goodman-Bacon \(2019\)](#), the “negative weight problem” associated with β arises when treatment effects evolve over time. Thus, one may wonder if such problems would still be present when considering more general, dynamic specifications such as (3.3). [Sun and Abraham \(2020\)](#) shows that this is still the case as the β'_e s associated with (3.3) do not recover easy-to-interpret causal parameters and still generally suffer from the same sorts of “negative weighting problems.” In contrast to this, we provide a simple way to directly aggregate our group-time average treatment effects into average treatment effects across different lengths of exposure to the treatment.

3.1 Aggregations to Highlight Treatment Effect Heterogeneity

Next, we discuss several partial aggregations of the group-time average treatment effects in order to summarize different dimensions of treatment effect heterogeneity. Although there are additional possibilities, we focus our discussion below on how to answer three particular questions: (a) How does the effect of participating in the treatment vary with length of exposure to the treatment? (b) Do groups that are treated earlier have, on average, higher/lower average treatment effects than groups that are treated later? (c) What is the cumulative average treatment effect of the policy across all groups until some particular point in time? Throughout this section, to avoid notation clutter, we assume that units do not anticipate treatment, i.e., we consider the case where Assumption 3 holds with $\delta = 0$. We also assume that a “never treated” group is available.

3.1.1 How do average treatment effects vary with length of exposure to the treatment?

One of the most popular questions that arises in DiD setups with multiple time periods concerns treatment effect dynamics: How does the effect of participating in the treatment vary with length of exposure to the treatment? For instance, do average treatment effects increase/decrease with elapsed treatment time? Indeed, answering this type of question is often the main motivation for using the event study regression in (3.3), though, as we mentioned above, that sort of regression may not be suitable for such a task. In this section, we propose an aggregation scheme that is suitable to highlight treatment effect heterogeneity with respect to length of exposure to the treatment that does not suffer from the drawbacks associated